



EXCHEQUER

Smart India Hackathon 2026 · SIH26183 · Ministry of Home Affairs

Exchequer

A proposal for institutional support: tracing cryptocurrency fraud from a victim's wallet to the exchange that cashed it out

SUBMITTED BY	Sidhyarth, B.Tech, Alliance University (SIH entry: WiFiBandits)
REQUEST	Running costs, a faculty mentor, and a letter of introduction to a police cyber-crime cell
STATUS	Working software, hosted, tested, with a published validation study

3

blockchains traced

407

verified exchange wallets

234

automated tests

88

real wallets measured

1 In one page

Exchequer is a piece of software I have built for Smart India Hackathon 2026, for a problem statement set by the Ministry of Home Affairs. A victim of a cyber-fraud reports the cryptocurrency wallet address they paid into. Exchequer follows the money outward from that address, hop by hop across the public blockchain, flags the known laundering shapes it passes, identifies the cryptocurrency exchange where the funds finally landed, and produces a report an investigating officer can attach to a lawful request to that exchange. The exchange, which by law holds the customer's identity, is the only party that can turn an address into a person; Exchequer's job is to get the officer to the right address with the evidence in order.

The problem is not small. By the Ministry's own figures tabled in the Lok Sabha on 2 December 2025, Indian citizens reported losses of **Rs. 2,290 crore** to cyber fraud in 2022, **Rs. 7,465 crore** in 2023 and **Rs. 22,845 crore** in 2024, from 2.27 million complaints on the National Cyber Crime Reporting Portal in that year alone. Much of the proceeds leave the country as stablecoins: Chainalysis measured stablecoins at 84% of illicit cryptocurrency volume in 2025, and the UN Office on Drugs and Crime describes USDT on the Tron blockchain as the “preferred choice” of the South-East Asian cyber-fraud operations that target Indian victims.

The software exists and works today. It traces Ethereum, BNB Smart Chain and Tron; it matches against 407 exchange wallets that were each verified on-chain before being accepted; it screens every address against the US Treasury sanctions list; it hashes every piece of evidence so a report can be checked in court; it has 234 automated tests that run on every change; and it is deployed on a public URL for judges to use. I measured my own detection rules against 88 real wallets and published the results, including the inconvenient ones.

What I am asking for. The University's support for the running costs of the project — paid data-provider tiers and hosting, and travel to the SIH finale — a faculty mentor from the department, and a letter of introduction so that I can put the tool in front of a working cyber-crime cell. Section 6 sets out what each would enable; I would bring a line-item budget to a follow-up meeting.

2 The problem, precisely

A cyber-fraud complaint in India starts at the state police or the National Cyber Crime Reporting Portal. When the victim paid in cryptocurrency, the officer holds one thing: a wallet address. Every transaction that address ever made is public and permanent, but reading that history by hand means opening a block explorer, copying each outgoing transfer, opening the recipient, and repeating, across dozens of wallets that the launderer created precisely to make this tedious. The commercial tools that automate it (Chainalysis, TRM Labs, Elliptic) cost lakhs per seat per year, are licensed to central agencies, and produce scores from models that cannot be cross-examined.

What the officer actually needs is narrower than what those tools sell. The Prevention of Money-Laundering Act made cryptocurrency exchanges *reporting entities* on 7 March 2023: they must register with FIU-IND, keep records that “enable it to reconstruct individual transactions” and identify the customer (section 12), and hand them over on a summons under section 50 or a police notice. So the whole investigative question reduces to: **which exchange, which deposit address, which transaction hashes, what amounts, when.** That is exactly, and only, what Exchequer produces.

Why a student project can do this at all

Because the data is public. Every transfer on Ethereum, BSC and Tron can be read from free explorer APIs, and the exchanges' own hot-wallet addresses are published by those same explorers. The engineering problem is to walk the graph efficiently within a free API's rate limit, to match honestly against published labels, to refuse to guess, and to write down what the tool did in a form that survives a courtroom. None of that needs proprietary data. It needs care, and it needs tests.

3 What I have built

The repository holds roughly 9,700 lines of application code across a Python (FastAPI) backend and a React frontend, written between 3 and 18 September 2026, with 65 commits and a continuous-integration pipeline that runs the full test suite and the frontend build on every push. The accompanying research report describes each part in detail; the summary is:

Capability	State on 19 September 2026
Chains and assets	Ethereum (ETH, USDT, USDC), BNB Smart Chain (BNB, USDT, USDC), Tron (TRX, USDT). Forward (where did it go) and reverse (who funded it) traces.
Exchange attribution	407 verified wallets across 18 exchanges, including CoinDCX. Each was imported by script from the explorer's own label page and checked on-chain before entry; four mislabelled contracts were rejected by that check.
Deposit-address inference	Names the unlabelled wallet one hop before the exchange when its entire history is sweeps to one labelled wallet, scores it lower, and says what would confirm it. Fired on 0 of 48 control wallets.
Sanctions screening	Every address checked against the OFAC SDN list of 4 September 2026 (124 Ethereum, 282 Tron, 1 BSC addresses). A trace stops at a mixer by design.
Pattern detection	Peel chain and amount split as fixed arithmetic rules that print their thresholds. Measured on 40 documented-illicit and 48 ordinary wallets; results published in the README.
Time rule	Only transfers made at or after the traced funds arrived are followed, so a busy wallet's unrelated prior history is never reported as the victim's money.
Swap detection	A swap at a labelled DEX router is read from the transaction receipt and reported with the output asset and where to resume.
Cross-case correlation	Finds intermediaries shared by traces from different complaints: on 15 September three Phish/Hack-labelled wallets were found to share 61 intermediaries, four of them inferred Binance deposit addresses.
Evidence integrity	Every provider response is SHA-256 hashed with its request and time; the report carries the manifest and ends with a content hash over itself.
Confidence score	A plain weighted sum of three printed inputs. No model, no clustering. Null, not zero, when nothing was attributed.
Interface	A case view with a hop-layered flow diagram, a transfers feed, a sealed text/JSON report, a projector "present mode", and a live ledger backdrop.
Quality	234 tests, green CI, disk-persistent response cache (flagship trace 43 s cold, 0.12 s cached), Docker image deployed on Railway.

Three commitments run through the design and I would ask that any support keep them: every attribution is an exact match against a file a human can open; every rule prints the numbers it compared; and where my sources said less than I did, I changed my claim, not the source.

4 Why this matters to the University

- **A national problem statement, in the open.** SIH26183 is set by the Ministry of Home Affairs. A working, tested, hosted submission with published validation is the kind of result the department can point to, whatever the competition outcome.
- **Publishable work.** The validation study (two pattern rules, 88 real wallets, a 68-point threshold grid) and the deposit-address inference rule are, to my knowledge, not written up anywhere in the open literature at this level of reproducibility. With a mentor I would like to write it as a short paper.
- **A relationship with law enforcement.** The tool is only finished when an officer has used it on a real complaint. A letter from the University opens that door in a way a student on his own cannot.
- **A foundation for further projects.** The upgrade path (Bitcoin, more chains, PDF reports, a Postgres backend) is a set of well-scoped tasks that later batches can take up as course or capstone work.

5 Plan for the next six months

When	Milestone	Outcome
Oct 2026	SIH 2026 next round (same judges as the internal round)	Demo on the hosted instance; four rehearsed cases; Q&A; prepared from JUDGE_QA.md
Oct–Nov 2026	Paid data tiers	Etherscan Lite unlocks Polygon, Arbitrum, Optimism, Base and Avalanche through code already present; Tron and BSC tiers remove the per-second ceiling that makes a cold trace take 20–40 s
Nov 2026	PDF report renderer	The text report already carries the evidence manifest; the PDF is a renderer, not new analysis
Nov–Dec 2026	Pilot with a cyber-crime cell	Five real complaints traced with an officer present; findings written up; the tool changed where it fell short
Dec 2026–Jan 2027	Bitcoin adapter	A different transaction model (UTXO); its own module, first item on the upgrade path
Jan–Mar 2027	Postgres, concurrency guard, frontend tests	Multi-user readiness for a shared instance
Mar 2027	Short paper	The validation study and the deposit-address rule, with the mentor as co-author

6 What support would enable

The project has run so far on free API keys, a free hosting trial and my own time. Each of the items below is modest; none is a salary, a proprietary data licence or a commercial analytics subscription. Everything the tool matches against is public and stays public. I would rather agree the direction first and bring exact figures to a second meeting.

Item	What it enables
Etherscan Lite plan	Five more EVM chains (Polygon, Arbitrum, Optimism, Base, Avalanche) through code already present, and a higher rate limit

Item	What it enables
TronGrid and NodeReal paid tiers	Removes the free-tier per-second ceiling that makes a cold trace take 20–40 s on Tron and BSC
Hosting	A persistent instance with a volume, so the demo is always up rather than brought up two days before a round
Domain and e-mail	A stable address to give an officer or a judge
SIH finale travel and stay	Only if the finale is held outside Bengaluru and SIH does not cover it
Printing and demo material	A poster and printed sample reports for the room

7 What I ask for beyond costs

1. **A faculty mentor** from the department, ideally with an interest in security, networks or data systems, for a fortnightly review and as co-author on the paper.
2. **A letter of introduction** on University letterhead to a police cyber-crime cell (Bengaluru City or Karnataka CID), asking for a supervised pilot on closed or ongoing cases at the officer's discretion.
3. **A demonstration slot** to the department, so that other students can see the problem and the upgrade path.
4. **Recognition of the work** as project credit, if the department's rules allow it.

8 Accountability

- A one-page progress note to the Director at the end of every month, with the commit log and the test count.
- The repository stays under version control with continuous integration; the Director or mentor is given read access.
- The hosted instance is available for the department to try at any time; the demo sign-in is admin / admin and is not access control.
- Any change to a detection rule requires two-sided tests and a written explanation in the README before it ships. This is already how the project works and I would keep it.

9 About me

I am Sidhyarth, a first-year B.Tech student at Alliance University. I built Exchequer on my own for Smart India Hackathon 2026 — the backend, the frontend, the label pipeline, the validation study and the documentation — and I present it. I would be glad to demonstrate the software live in the Director's office at any time; a trace of a real fraud address takes under a minute.

Sidhyarth
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Attached: *Exchequer — Research Foundations and Work to Date*, the companion report that explains the cryptocurrency concepts involved, the published evidence the project rests on, and everything built so far.